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Non-uniqueness in surface-wave inversion and consequences on seismic site response analyses

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ABSTRACT

Surface-wave tests are based on the solution of an inverse problem for shear-wave velocity profile identification from the experimentally measured dispersion curve. The main criticisms for these testing methodologies are related to the inverse problem solution and arise from the possible equivalence of different shear-wave velocity profiles. In this paper, some implications of solution non-uniqueness for seismic response studies are investigated using both numerical simulations and experimental data. A Monte Carlo approach for the inversion problem has been used to obtain a set of equivalent shear-wave velocity models. This selection is based on a statistical test which takes into account both data uncertainty and model parameterization. This set of solutions (i.e., soil profiles) is then used to evaluate the seismic response with a conventional one-dimensional analysis. It is shown that equivalent profiles with respect to surface-wave testing are equivalent also with respect to site amplification, thus countering the criticism related to inversion uncertainty for the engineering use of surface-wave tests.
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1. Introduction

The small-strain shear modulus, which is a fundamental parameter for seismic site response analysis, is strongly affected by disturbance induced by sampling and reconsolidation in laboratory: it is therefore necessary to use in-situ seismic methods to obtain a reasonable estimate [1]. Non-invasive seismic tests have become more and more popular in this respect, because they are cost and time effective options for testing large areas and they provide global information on the behavior of the soil deposit. Among them, surface-wave tests are widely used, also because they do not suffer from theoretical and practical limitations that affect body-wave techniques (e.g., hidden or inverse layers in SH-wave refraction surveys).

Surface-wave tests are based on the property of geometrical dispersion which makes the velocity of propagation of Rayleigh waves frequency dependent in vertically heterogeneous media. The experimental dispersion curve, which can be extracted from field data using a variety of processing techniques [2,3], is used for the solution of an inverse problem aimed at the identification of the soil model parameters. In particular, the soil deposit is usually modeled as a stack of homogeneous linear elastic layers and the parameters to be identified are the layer thicknesses and shear-wave velocities. Poisson ratios and density of each layer are usually assigned a priori because of the small sensitivity of the Rayleigh wave dispersion curve to these parameters [4]. Thus typical values are assumed even if the evaluation of such parameters for soils in their natural state is not trivial.

Since surface-wave tests are based on the solution of an inverse problem, the major criticism in their respect is related to the non-uniqueness of the solution and to the equivalence problem. These drawbacks affect most of the geophysical inverse problems which, for their nature, are often ill-posed from the mathematical point of view. Indeed it is not possible to prove the uniqueness of their solution [5]. The equivalence problem is, for example, well recognized in vertical electric soundings where the experimentally measured apparent resistivity curve has very little sensitivity to resistivity and thickness variations which do not modify the longitudinal conductance or the transverse resistance of a layer.

In surface-wave inversion, the equivalence of different dispersion curves concerns mainly the low sensitivity of the dispersion curve to model parameter variations for complex models, which leads to low-resolution results and consequent difficulties in defining the optimum model parameterization for the inversion process. This inherent ambiguity means that several different

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profiles can be associated to numerical dispersion curves of Rayleigh waves having an equally good agreement with the experimental one. It is possible, therefore, to identify a number of sets of soil parameters which are intrinsically equivalent, i.e., their associated misfit is small if compared to the experimental uncertainty caused by measurements and processing techniques adopted for the estimation of the experimental dispersion curve. This causes a certain level of ambiguity in the final shear-wave velocity profile, which is used for modeling the site seismic response.

Different approaches have been chosen, and are commonly used, to invert the experimental dispersion curve. Most of them are based on deterministic inversion techniques (i.e., linearized inversion techniques), with some differences in the data concerned, the model parameters, the computation of the partial derivatives, the inversion algorithms, the use of constraints, etc. [6]. Local-search (LS) methods are based on local derivative approximations [7] evaluated by the best data fitting model, hence they become less accurate as the data–model relationship becomes strongly non-linear. Shapiro [8] showed that the solutions obtained from classical surface-wave inversion schemes (e.g., damped least-square algorithms) are too restrictive and uncertainties are not correctly estimated. The shear-wave velocity profile selected by LS methods is only one of the possible solutions. The result is very sensitive to the initial model and the inversion process can easily be biased by wrong choices in terms of model parameterization that lead the solution into local minima [9–11].

Nevertheless, it is worthwhile to consider that since the dispersion curve is associated to the global dynamic behavior of the site and a soil model similar to the one adopted for the interpretation of surface-wave data (a stack of homogeneous isotropic layers) is used for several engineering applications, the equivalence problem is likely to have no severe consequences. Looking specifically at seismic site response studies, it is possible to state intuitively that if several soil profiles are equivalent in terms of surface-wave propagation, they are likely to be equivalent also in terms of the seismic site amplification, if the latter is evaluated using a 1D-layered model.

This aspect is systematically explored in the present work using a Monte Carlo approach for the inversion process. The space of model parameters is randomly sampled and a set of profiles is then selected through a statistical test. The selected profiles are equivalent with respect to the experimental dispersion curve and its uncertainties. In principle, if the sampling of model parameter space is adequate and the chosen parameterization is the same, any deterministic solution to the same inverse problem should fall within this set of selected profiles.

Equivalent profiles are subsequently used for 1D numerical simulations of seismic site response in order to assess the influence of uncertainty and non-uniqueness of the inversion solution. A synthetic example, for which the "exact" solution is available, and two real case histories are reported.

2. Surface-wave tests

A variety of approaches exists to determine experimentally the dispersion curve. It is possible to distinguish, in particular, between active surface-wave tests, in which the waves are on-purpose generated using a seismic source [12–14], and passive surface-wave tests, which are based on the analysis of microtremors [15–17]. The main difference in terms of results between active and passive surface-wave tests is related to the different frequency band in which the information can be retrieved. Indeed, while in active tests it is usually very easy to generate and detect high frequency components, microtremors are typically rich in energy in the low frequency band. This affects the investigation depth (related to the available information at low frequency) and the spatial resolution close to the ground surface (related to high frequency band data).

An adequate profile characterization, which requires the evaluation of the deep structure and the location of the seismic bedrock together with the shear-wave velocity profile of the sediments, should include both testing methodologies; in this respect, the combined use of passive and active methods has been suggested to improve both resolution and depth of investigation [18–20].

3. Monte Carlo inversion

In the present study, a Monte Carlo (MC) procedure has been used for the inversion of experimental surface-wave dispersion curves (a detailed description of the procedure can be found in [6]). With such an approach, it is possible to obtain a set of equivalent shear-wave velocity profiles according to available information and accounting for data quality and model parameterization. This inversion algorithm has been designed to be particularly efficient thanks to the introduction of scale properties of the dispersion curves [21]. These properties are linked to the scaling of the modal solution with the wavelength. If model parameters are scaled, the corresponding modal dispersion curve scales accordingly: in particular, both phase velocities and frequencies scale if all the layer velocities are scaled, while only the frequencies scale if all the layer thicknesses are scaled [3].

A set of random models is generated after defining the number of layers and the starting upper and lower boundary for each model parameter (shear-wave velocity and thickness of each layer). This set of models is then used to calculate the associated synthetic dispersion curve, assumed to be the fundamental mode of pseudo–Rayleigh wave propagation, using a forward problem algorithm based on the transfer matrix approach [22–24]. Each of the obtained synthetic dispersion curves is then shifted as close as possible to the experimental one. This operation supplies the scale factor by which the corresponding model is updated (back applying the scale factor to the randomly generated model). The obtained scaled models are then used in the inversion and the misfit is hence calculated between the experimental dispersion curve and the shifted ones. This procedure concentrates the sampling in the high probability density zones of the model parameter space [6] and allows for a significant optimization of the process, reducing the number of required simulations (usually around 10^4–10^5).

Assuming that the experimental dispersion curve is affected by a Gaussian error [25] with a known standard deviation, a misfit function with the structure of a chi-square can be used:

$$S = \chi^2 = \frac{\sum_{i=0}^{l}(v_{ti} - v_{ei})^2 W_i \sigma_{ei}^{-2}}{l - (2n - 1)} \tag{1}$$

where v_t and v_e are the theoretical and experimental phase velocity vectors, respectively; W is a weight that can be used to account for the sampling issues and which is taken equal to the unit vector in the present work; σ_e is the data uncertainties vector; l is the number of data points in the dispersion curve and n is the number of layers of the model (only shear-wave velocity and thickness for each layer are considered unknown, while the densities and Poisson ratios are fixed a priori).

The denominator of Eq. (1) contains, in this way, the degree of freedom of the problem which is given by the difference between the amount of experimental information and the number of unknowns.

A statistical one-tailed Fisher test can be applied, knowing that the ratio between chi-square variables follows a Fisher distribution (see for instance [26]). This procedure selects the set of acceptable velocity models that represents the final result of the inversion according to a fixed level of confidence α. In this way, it is possible to account for both the data quality and the dimensionality of the problem. Indeed, the experimental uncertainties are included in the χ^2 and the statistical test is ruled by the degrees of freedom of the Fisher distribution. The higher the confidence, the wider will be the set of selected profiles [27].

This inversion approach allows one to get a "picture" of non-uniqueness of the solution in surface-wave analysis and supplies a tool to assess the consequences on seismic response studies or other engineering applications.

4. Synthetic test

A synthetic dataset has been generated to assess the consequences of non-uniqueness with respect to the "exact" solution. First, we applied the MC inversion to the data and obtained a set of equivalent models; then, we simulated the seismic response of this set of models. For this preliminary study, we chose a reference velocity profile composed of two layers over the seismic bedrock (Table 1). The numerical dispersion curve has been generated for the given "real" reference velocity profile through the same forward solver implemented in the inversion algorithm; then, it has been perturbed with Gaussian noise and used in the inversion process. The Gaussian noise is aimed to simulate real data acquisition and it has, therefore, been designed according to the typical pattern of uncertainties in real data [25,28]. The coefficient of variation of the "experimental" dispersion curve varies linearly with frequency from 7% at the lowest frequency to 4% at the highest frequency. In the following, the dispersion curve with noise relative to the reference model will be defined as reference dispersion curve. The reference curve was inverted using a confidence level α equal to 1%: the result of the inversion is, therefore, a set of models that fulfill the condition of containing all the equivalent models among the whole randomly generated model set at 99% probability.

Table 1
Synthetic velocity profile compared to the results of the MC inversion.

Synthetic Soil Profile		Monte Carlo inversion			
		Mean value	Standard deviation	Mean CoV %	Maximum CoV %
Thickness (m)	5	4.8	0.4	8.9	9.9
	10	9.1	1.7	18.6	21.2
Vs (m/s)	200	198.1	2.1	1.1	1.1
	400	382.2	38.0	10.0	11.0
	1000	982.9	71.8	7.3	7.8

The result of the inversion is a set of six shear-wave velocity profiles (Fig. 1a). The synthetic dispersion curve relative to each of the selected profiles is compared with the reference curve: all of them show a very good fit (Fig. 1b). It is well clear from Fig. 1b that any of the selected profiles can be considered as an "equally good" solution of the inverse problem, since, considering data uncertainty, there is no way to discriminate among them the most consistent result. A deterministic inversion with a local-search algorithm of the same dispersion curve could lead to any of the six shear-wave velocity profiles selected by the MC algorithm.

In Fig. 1 a representation based on the misfit (Eq. (1)) is adopted, for both dispersion curves and corresponding velocity profiles; the darkest color corresponds to the curve having the lowest misfit and better approximation to the reference dispersion curve. The same color used for each curve is used to represent the model associated to that curve, so darkest model corresponds to the best fitting curve. The same representation is used consistently throughout the paper, associating also the results of the seismic site response simulations from a given velocity profile with the misfit value of its dispersion curve.

As shown in Fig. 1b, the best fitting profile is very close to the reference profile; nevertheless, the other solutions are also associated to dispersion curves sufficiently close to the reference one. Table 1 reports a statistical analysis of the results of the MC inversion compared to the reference velocity profile. Data are also presented in terms of coefficient of variation (CoV), defined as the ratio between standard deviation and the exact value of each parameter (thickness and velocity of each layer) within the six different soil profiles. We will also use this parameter to interpret the result of the seismic response analysis.

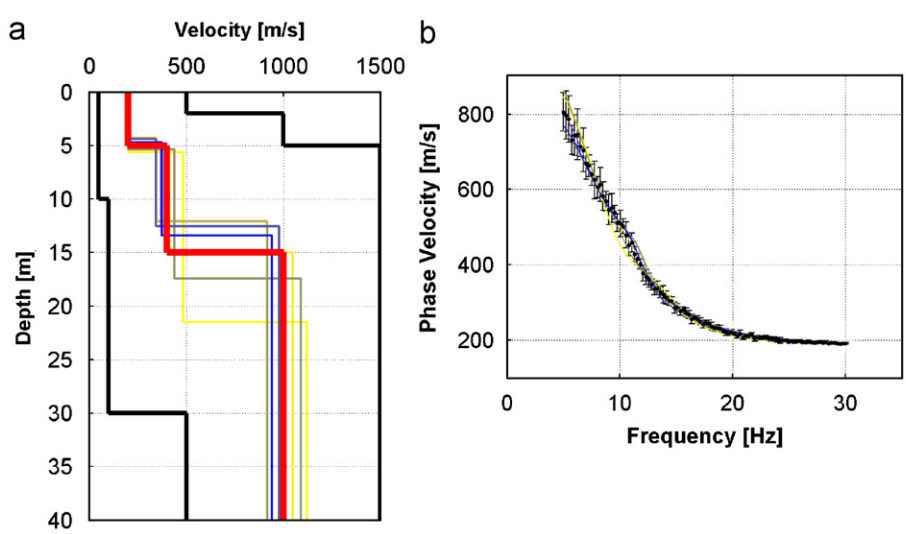

Fig. 1. Synthetic example: (a) shear wave velocity profiles from the MC inversion compared to the reference velocity profile (thick red line); the outer profiles in black are the boundaries for the random generation of the population of velocity profiles; (b) dispersion curves associated to the solutions of the MC inversion compared to the reference dispersion curve (black dots).

The mean values of the identified parameters for the six profiles are quite close to the reference values, with uncertainty increasing with depth. It is noteworthy that the differences between the resulting profiles are significant especially for the position of the second interface, with a maximum value of the CoV of 21.1%, whereas the differences in the top portion of the models are slightly reduced. These results are consistent with the lower experimental uncertainty of surface-wave tests at high frequencies that leads to higher resolution close to the ground surface.

Since, in principle, any of the profiles in Fig. 1a can be considered as a possible solution from a deterministic point of view, the error in the position of the second interface could be as high as 50%. In order to investigate the consequences of the uncertainty in the shear-wave velocity profile, we simulated the seismic site response for each velocity profile using the code Shake91 [29]. The results of the numerical simulations were then compared.

Initially, we performed a linear elastic analysis in order to assess the problem of non-uniqueness avoiding influence on the results from the choice of stress–strain relationships and seismic input. In Fig. 2a, the amplification functions of the equivalent profiles are presented and compared to the one obtained from the reference profile. In Fig. 2b, the comparison is reported in terms of response spectra for a given accelerogram (a record from the Umbria earthquake of 1997).

These results clearly show that the differences between the seismic responses obtained with the six equivalent profiles are negligible; hence, we can deduce that equivalent profiles in terms of surface-wave inversion are equivalent also in terms of seismic site response. In particular, the first resonance peak of the amplification function is very well identified with all the equivalent profiles, both in terms of frequency and amplification value. The amplification functions are practically coincident for the frequency band below 10 Hz, which is relevant for most earthquakes, and the relative difference on the value of the main resonance peak among the various profiles is of the order of 3–5%. Also the differences in terms of response spectra (Fig. 2b) are negligible, so that it is possible to state that the profiles obtained with surface-wave inversion are equally good from the point of view of site response prediction.

In Table 2, the mean and maximum values of the CoV evaluated at each frequency of the response spectra for the six different profiles are reported. It is noteworthy that the CoV in the spectral ordinates is reduced with respect to the CoVs for parameters of the soil profile (Table 1). In particular, referring to the high value of the CoV in the evaluation of the second layer thickness (21.1%), the maximum uncertainty in the spectral ordinates is three times lower. This means that the uncertainty in the determination of the deeper portion of the velocity profile is attenuated in the evaluation of the seismic site response. Moreover, there is no correspondence between minimum misfit on the dispersion curve and best estimate of the seismic site response. Indeed profile #6, which is the highest misfit profile in the inversion, is the best fitting profile from the point of view of seismic response.

In order to confirm the above findings with more realistic analyses, the evaluation of site response has been repeated with a linear-equivalent approach. The normalized modulus reduction and material damping curves have been selected among available standard curves from the literature. In particular, the two top layers have been assigned degradation curves for sands [30], whereas the rock half-space has been considered linear elastic.

Since, in this case, the final result is strongly dependent on the input motion assigned at the outcrop, the analyses have been performed scaling the same record from the Umbria 1997

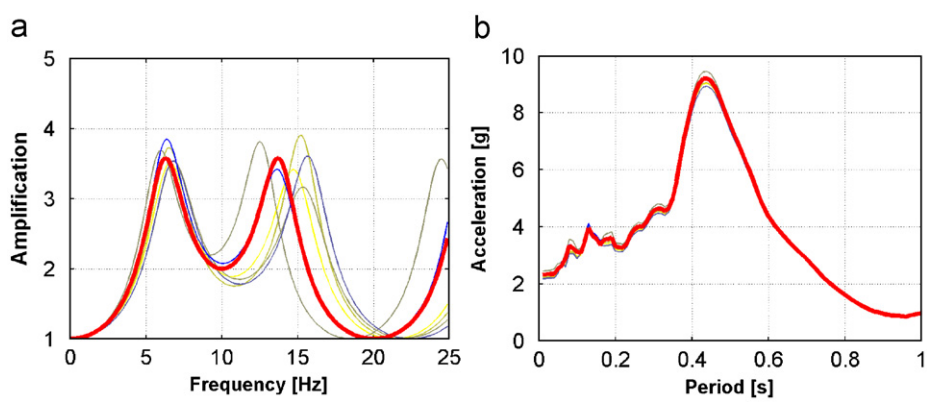

Fig. 2. Synthetic example—elastic analysis: comparison of results obtained with the equivalent velocity profiles from the MC inversion and the reference profile (thick red line): (a) amplification functions; (b) response spectra (record from the 1997 Umbria earthquake).

Table 2
Synthetic example—seismic site response of the equivalent velocity profiles for elastic and equivalent linear elastic analyses (record from the 1997 Umbria earthquake).

Model	Seismic zone	Response spectra		PGA (g)		
		Mean CoV %	Maximum CoV %	"Exact"	Mean	Standard deviation
Elastic	–	1.7	7.5	2.32	2.27	0.11
Equivalent-linear-elastic	1	3.8	6.8	0.57	0.55	0.03
	2	3.2	6.9	0.38	0.37	0.02
	3	2.6	6.4	0.21	0.21	0.01

earthquake to 0.15, 0.25 and 0.35 g, which are reference values for peak ground acceleration (PGA) on rock outcrop in Italy. The response spectra on the ground surface for different profiles, compared with the one obtained for the reference profile, are reported in Fig. 3. The CoV values are reported in Table 2 and they are of the same order of magnitude as the ones obtained in the elastic analysis. The PGAs for the set of equivalent profiles are also compared to the ones obtained from the reference velocity profile, showing good accuracy (mean values) and reduced overall uncertainty (standard deviations).

The obtained results confirm the conclusions drawn from the linear elastic analysis, i.e., the shear-wave velocity profiles retrieved by surface-wave inversion produce a site response estimation which is equivalent to the one obtained using the reference (i.e., the exact) profile.

The scaling of the same accelerogram for different values of PGA is, however, not fully realistic because the frequency content and many other motion parameters are quite different for strong and weak ground motions. Hence, it has been decided to complement the study on the synthetic example, by repeating the site response analysis with a more consistent set of input motions. In particular, a set of real earthquake records have been selected as the input motion [31]. The set is composed of seven accelerograms which are compatible with the design spectrum of Eurocode 8 [32], meaning that the average of their response spectra is close to the design response spectrum, according to the criteria described in Eurocode 8 (Fig. 4a). For each profile, the response spectra for the seven spectrum-compatible accelerograms have been evaluated and then averaged. Fig. 4b reports the comparison between the average spectrum for each of the profiles from surface-wave inversion compared to the results obtained for the reference profile. Again the results show only minor differences, whereas the shape and absolute values of the response spectra are very similar.

The results of simulations in terms of CoV are reported in Table 3, where a maximum value of 10% and a mean value of 2.5% for both sets of accelerograms can be noted. Also in this case, the "exact" response spectrum is within the range associated to the six equivalent profiles. This fact has a noticeable correspondence in the mean PGA values, which are in perfect agreement with the "exact" ones with low uncertainty.

From the different analyses conducted on this benchmark synthetic case, it can be clearly stated that the equivalent profiles in terms of dispersion curve inversion selected from the MC algorithm have a seismic response that, both in terms of response spectra and of amplification factors, fall within 5–10% of the "exact" one. Therefore, equivalent profiles in SW inversion are also equivalent in seismic site response terms.

5. Case histories

Two case histories are presented in order to evaluate the consequences of solution non-uniqueness on real data. For one of

Table 3
Synthetic example—seismic site response of the equivalent velocity profiles for the set of spectrum-compatible accelerograms.

Response spectra		PGA (g)		
Mean CoV %	Maximum CoV %	"Exact"	Mean	Standard deviation
2.5	10.3	0.63	0.63	0.02

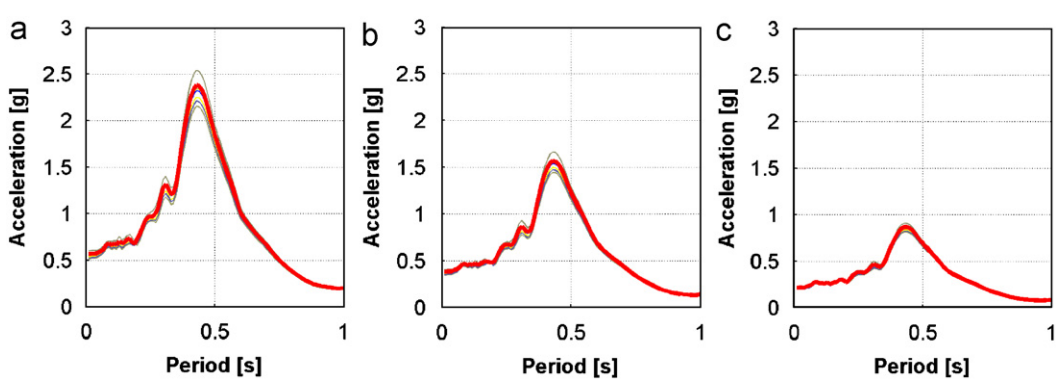

Fig. 3. Synthetic example—equivalent linear elastic analysis: comparison of response spectra from the equivalent velocity profiles from the MC inversion and the reference one (thick red line) for the Umbria 1997 earthquake record scaled for different values of outcrop PGA: (a) 0.35 g, (b) 0.25 g and (c) 0.15 g.

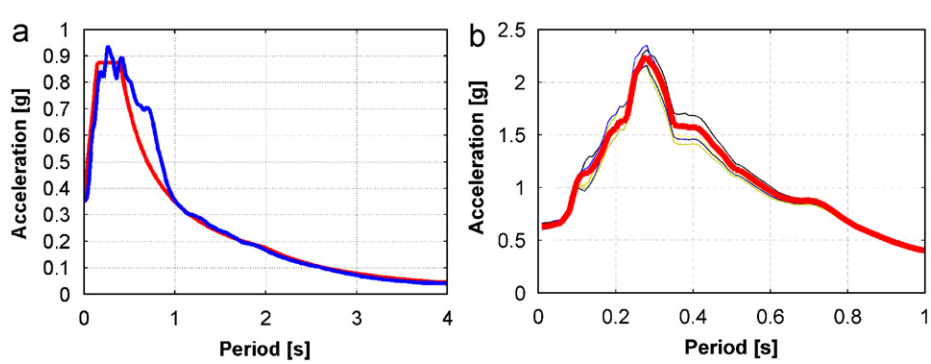

Fig. 4. (a) Average response spectra of the set of spectrum-compatible accelerograms compared to the design spectrum of EC8; (b) average response spectra of the set of spectrum-compatible accelerograms from the equivalent velocity profiles from MC inversion compared to the one relative to the reference velocity profile (thick red line).

the two sites, in which it was possible to obtain good quality samples, resonant column tests have been performed to obtain the normalized shear modulus reduction and material damping curves, which have been used for site response analysis. When undisturbed sampling was not feasible because of the nature of the sediments (gravelly and sandy layers), the curves have been taken from the literature [30].

The reference ground motions for the two sites have been selected on the basis of probabilistic seismic hazard studies. In particular, reference is made to the results available from a study conducted for the whole Italian territory [33] and to the simplified de-aggregation procedure proposed by Bommer et al. [34]. Using these parameters, real accelerograms recorded on rock sites have been selected from the European strong motion database [35]. These records have been then scaled to the reference PGA for each site.

As in the synthetic study, the seismic response of the sites has been evaluated using a linear-equivalent approach with independent runs of the numerical analysis for each of the profiles selected from the MC inversion, using the code Shake91 [29].

5.1. Torre Pellice site

The first case history is related to the Torre Pellice urban area, located in the Pellice Valley (Piedmont Region, Italy). The site presents an overburden mainly composed of a layer of fluvial sediments (compacted sands and gravels) with an expected thickness of 10–50 m over a lacustrine sediment layer (mainly composed by clays), down to the bedrock. The bedrock depth is expected to exceed 100 m in the central part of the valley. Undisturbed samples have been collected from the clayey layer to perform resonant column tests.

Surface-wave data used for the present survey have been extracted from a larger seismic survey [36]. In particular, a sub-set of 24 receivers has been considered. Data have been processed in the frequency–wavenumber (f–k) domain where the amplitude peaks correspond to the experimental dispersion curve [2]. The average values of phase velocities and their standard deviations have been obtained by processing different seismic records separately [36].

Fig. 5b shows all the numerical dispersion curves corresponding to the selected profiles in the MC inversion (reported in Fig. 5a) compared to the experimental one. It can be noticed that the data are quite noisy and that the uncertainties are significant in the low frequency band and above 20 Hz. The frequency band within which the dispersion curve was retrieved is limited at low frequency by the resonant frequency of geophones. The MC inversion result is a set of 81 equivalent velocity profiles selected among 5×10^4 randomly generated profiles. With respect to the synthetic example, the region of model parameter space delimitated by the result is wider, particularly in the deeper part of the model. This is a consequence of the relatively large uncertainty for this dataset. All the selected profiles present a velocity inversion (a softer layer within two stiff layers) which is related to the uppermost portion of the lacustrine sediments, as also confirmed by the down-hole measurements at a nearby site [36]. It is worth noticing that velocity inversions are a challenging target for seismic characterization, since they cannot be retrieved by seismic refraction. Furthermore, velocity inversions can also play a significant role in seismic site response. The synthetic dispersion curves relative to the selected profiles all lie within the uncertainties of the experimental one (Fig. 5b).

We run independent simulations of seismic site response for each of the selected shear-wave velocity profiles. The reference earthquake for Torre Pellice site is given by a magnitude 4.8 event at an epicentral distance 8 km, associated to a strike-slip rupture. The PGA on outcrop for the site is equal to 0.14 g for a return period of 475 years. Using these parameters, four real accelerograms have been selected for the site response study. The reference bedrock has been placed at 80 m from the ground surface on the basis of information retrieved from a combined P-wave reflection–refraction survey.

The results of the numerical analyses for one of the real accelerograms are reported in Fig. 6. In particular, Fig. 6a reports the amplification functions for different profiles, while Fig. 6b shows their response spectra. Again the color scale based on dispersion curve misfit is adopted, so that the darkest color corresponds to the profile relative to the best fitting dispersion curve in MC surface-wave inversion.

Similarly to the results obtained for the synthetic example, it can be observed that equivalent profiles in terms of surface-wave inversion are equivalent also in terms of site amplification, giving very similar response spectra on the ground surface (the average CoV of the spectra, calculated with respect to the mean values of the distribution, is 2% with a maximum absolute value of 8.9%). Moreover, the site amplification functions for different profiles are

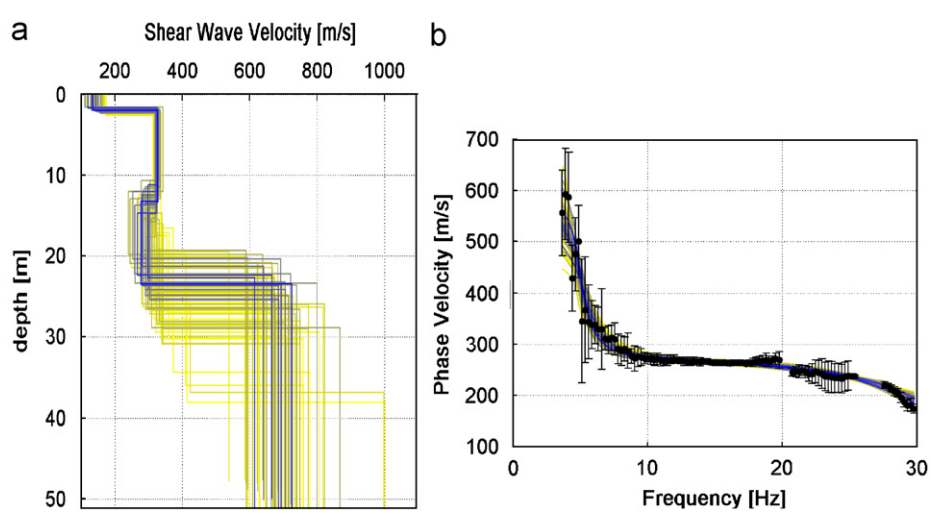

Fig. 5. Experimental results at Torre Pellice site—active surface wave tests: (a) shear-wave velocity profiles selected by MC inversion and (b) experimental and numerical dispersion curves.

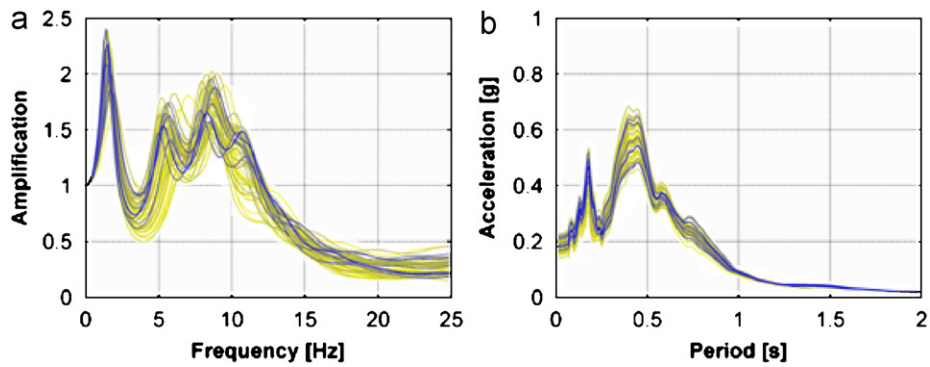

Fig. 6. Seismic site response at Torre Pellice site—(a) amplification functions and (b) response spectra for the equivalent velocity profiles of Fig. 7.

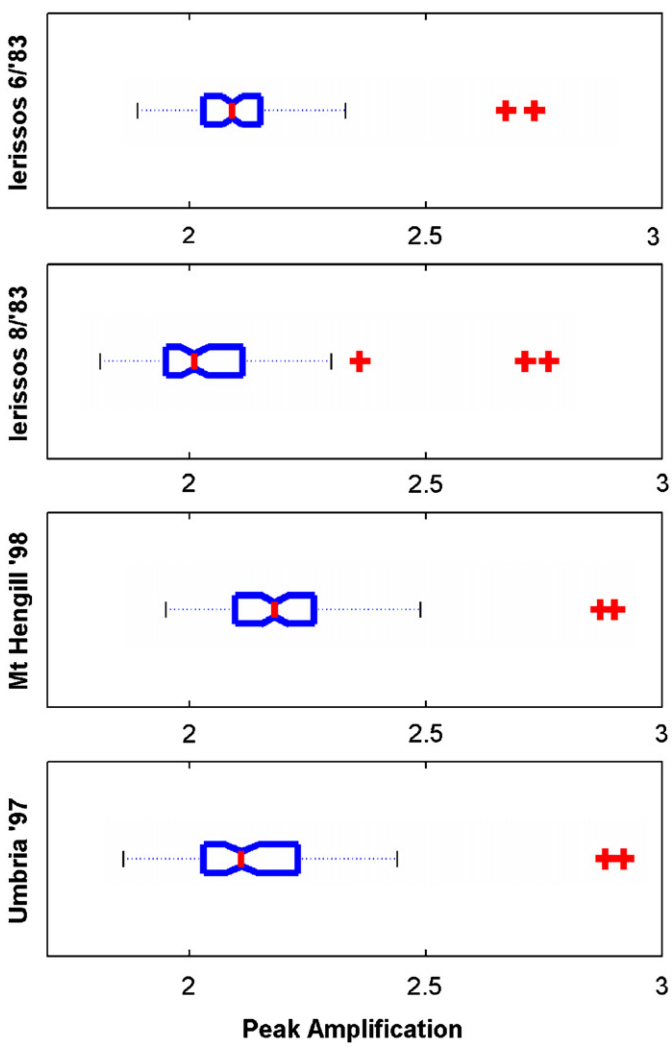

Fig. 7. Seismic site response at Torre Pellice site—box plots of the distributions of amplification maxima for different seismic inputs.

also very similar, particularly for the frequency band below 5 Hz, which is of major interest for site amplification studies.

Fig. 7 reports the results for the whole set of four accelerograms selected for the site in the form of box-plots, which synthetically describe the shape of the distribution, its central value, and spread of the peak in amplification functions retrieved for each seismic input from the different shear-wave velocity profiles.

Box-plots, which will also be used for the next case history, are characterized by three lines that identify, respectively, the lower quartile, median and upper quartile values of the distributions; the notches around the mean value represent an estimate of the uncertainty for box-to-box comparison (that means a confidence interval around the mean value) and the whisker lines, extending from each end of the boxes for 1.5 times the corresponding interquartile distance, are showing the extent of the rest of the data. Crosses are outliers data with values beyond the ends of the whiskers.

The statistical distributions of the peak amplification values obtained from the applications of the four selected accelerograms appear very similar with median values that show only minor variation between 2 and 2.2, and few outliers towards higher amplifications, evidencing the limited influence of the different input motions. Moreover, the dispersion of the results around median values, for each accelerogram, is very limited confirming that the uncertainty in the determination of the shear-wave velocity profiles does not have a strong influence in the evaluation of the main peak in the amplification function.

Fig. 8 reports the results obtained in terms of PGA. Also with respect to this parameter, the results consistently show a limited scale of variation (maximum CoV of 11.8%, with values of median peak acceleration ranging from 0.19 to 0.23 g).

5.2. La Salle site

The second case history refers to a site located in La Salle (Valle d'Aosta Region) on a wide alluvial fan. A borehole logged down to 50 m from the ground surface for the execution of a Down-Hole test reports layers of gravelly sands and silty sands with gravel. Laboratory tests were not feasible at this site, because the nature of the soil deposit prevented undisturbed sampling.

Surface-wave tests have been performed at this site using both active and passive methods [20]. The active data have been collected using a linear array of 48 vertical geophones (4.5 Hz), with 2 m spacing and a sledge-hammer at 2 m offset from the first geophone as a source. The acquisition of microtremors has been carried out using a 75 m circular array with 12 evenly-spaced vertical geophones (2 Hz). Both active and passive data were processed through f–k approach to retrieve the pseudo-Rayleigh wave dispersion curves. In the case of passive data, a beam-former technique [37] was adopted to handle the 2D array geometry. Average values and standard deviations of the experimental dispersion curves have been obtained processing several acquisitions with equal setup. Active data supplied information in the frequency band from 14 to 40 Hz, while the dispersion curves retrieved by passive data ranged from 2 to 18 Hz. The very good agreement of the two datasets in the common frequency band

allowed merging the data and obtaining a unique dispersion curve over a wider frequency band [20].

The availability of the two experimental datasets allowed assessing the improvement that can be achieved by merging active and passive data with respect to a single dataset. Indeed, we inverted both the dispersion curves: the one retrieved by the merging of active and passive data and the one obtained just from passive data. We compared the results in terms of retrieved velocity profiles and in terms of seismic site response analysis. This double analysis has the scope of assessing the consequences of lack of information in the high frequency range, which is the case when only passive data are collected.

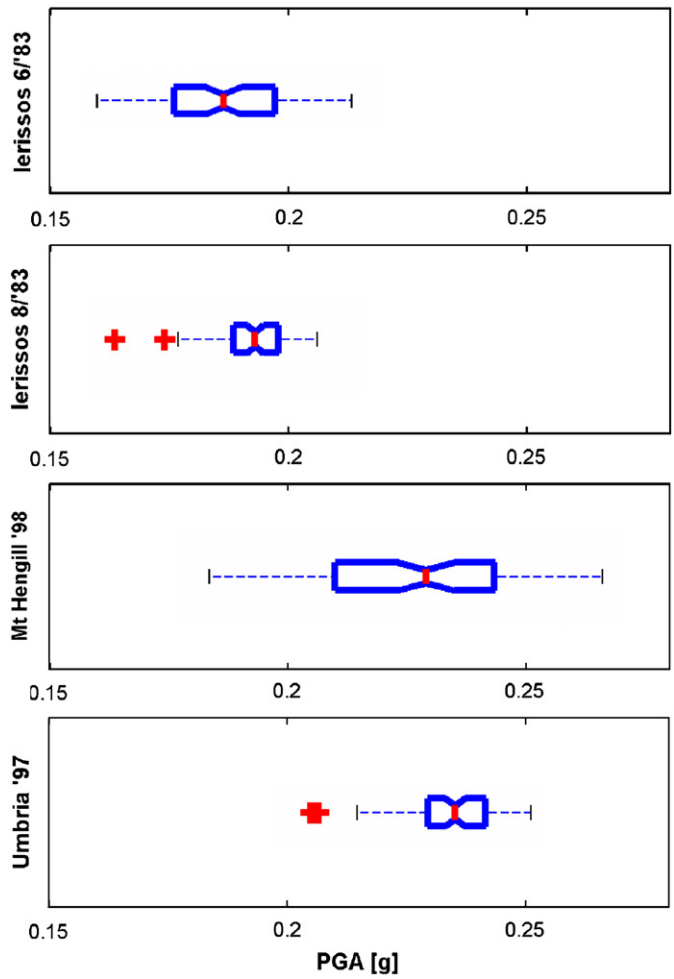

Fig. 8. Seismic site response at Torre Pellice site—box plots of the distributions of PGA for different seismic inputs.

Figs. 9 and 10 report the results of the Monte Carlo inversion for the two datasets. The combination of active and passive data (Fig. 9) leads to a much higher resolution for shallow layers, thanks to the availability of high frequency data. Moreover, starting from a population of 10^5 randomly generated velocity profiles, the number of profiles selected by the statistical test for the same level of confidence is much lower for active-plus-passive data (16) than for the case in which only passive data are used (270). This is partly due to the fact that the misfit function is influenced by the standard deviation of the experimental data and this is much lower for active tests (see Fig. 9b), but depends also on the degree of freedom of the problem, which is related to the number of available data points (see Eq. (1)).

In general, it is possible to state that the experimental test in which only passive data are used leads to a much higher uncertainty in the velocity model estimation. This fact is clearly evident in Fig. 11 where the box-plots of velocity distributions for each layer are reported (results are normalized with respect to the median of each layer value) for the case of combined active and passive tests (Fig. 11a) and for passive tests only (Fig. 11b). The uncertainty in the determination of the velocity of each layer is generally much lower for the combined active and passive data.

The positive effect of active data is twofold. First of all, as expected, the inclusion of high frequency information drastically improves resolution and reduces the uncertainty at shallow depths. Secondly, this improvement of the velocity model induces also an improvement for deeper strata with a noticeable reduction of the range of variation for velocities. This can be clearly seen in the box-plots of Fig. 11a where, with respect to passive data (Fig. 11b), there is a reduction of the uncertainty in the velocity values and an increase in notch width, underlining the higher confidence in the determination of the parameter.

After inverting the dispersion curves, we performed seismic site response analysis for all the retrieved velocity profiles and we compared the results with particular regards to the difference

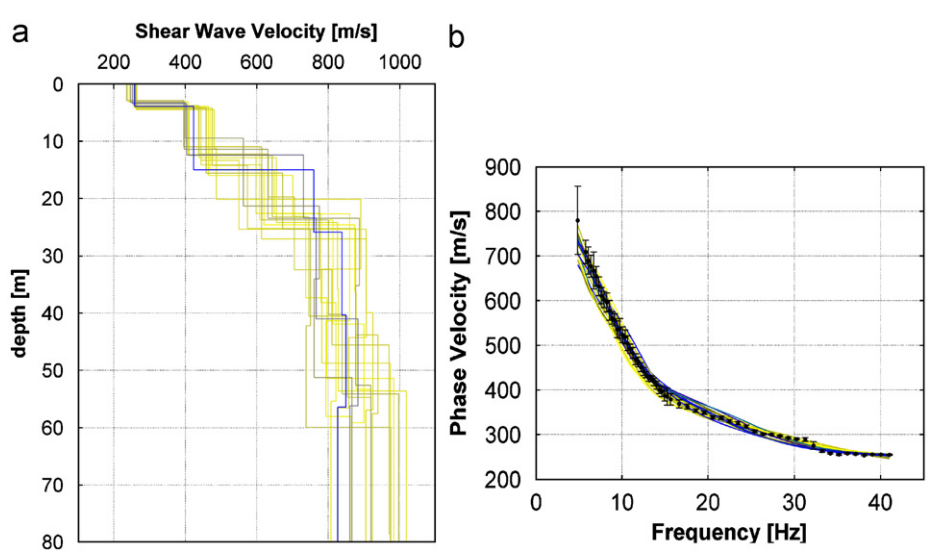

Fig. 9. Experimental results at La Salle site—combined active and passive surface wave: (a) shear-wave velocity profiles from the MC inversion and (b) dispersion curves associated to the solutions of the MC inversion compared to the experimental dispersion curve (black dots).

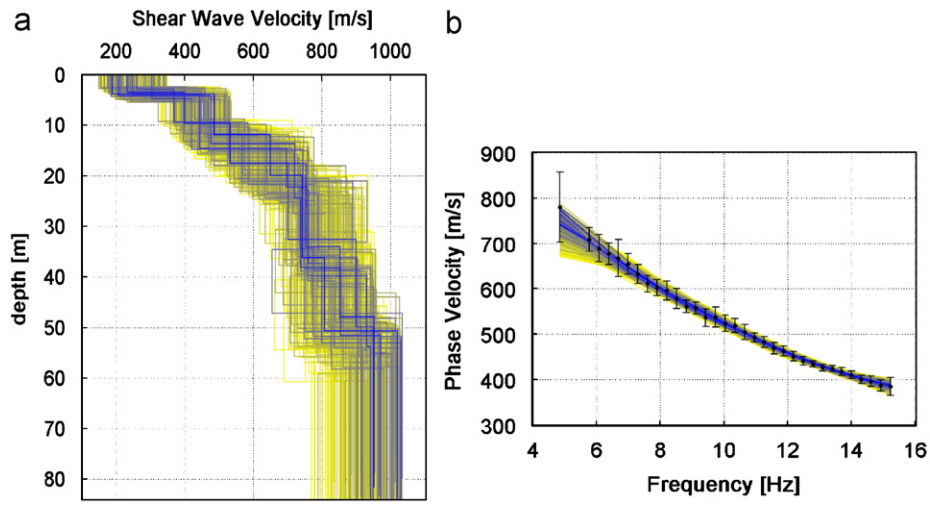

Fig. 10. Experimental results at La Salle site-passive surface wave: (a) shear wave velocity profiles from the MC inversion and (b) dispersion curves associated to the solutions of the MC inversion compared to the experimental dispersion curve (black dots).

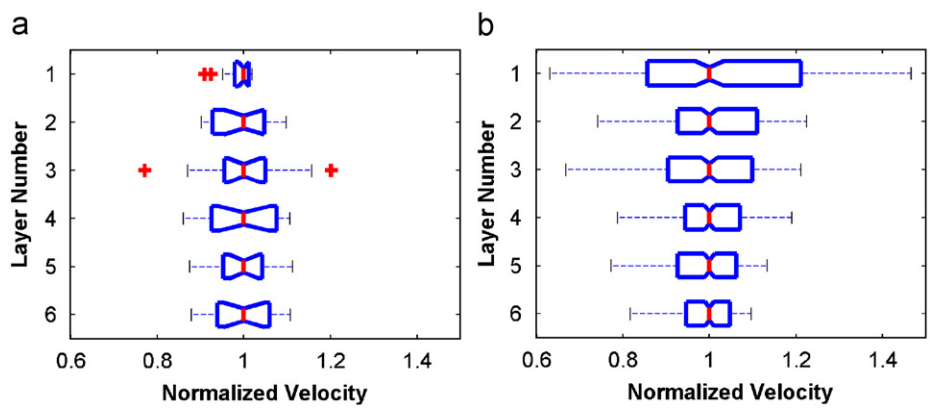

Fig. 11. Experimental results at La Salle site—box plots of the distributions of the normalized velocity for each layer: (a) combined active and passive tests and (b) passive tests.

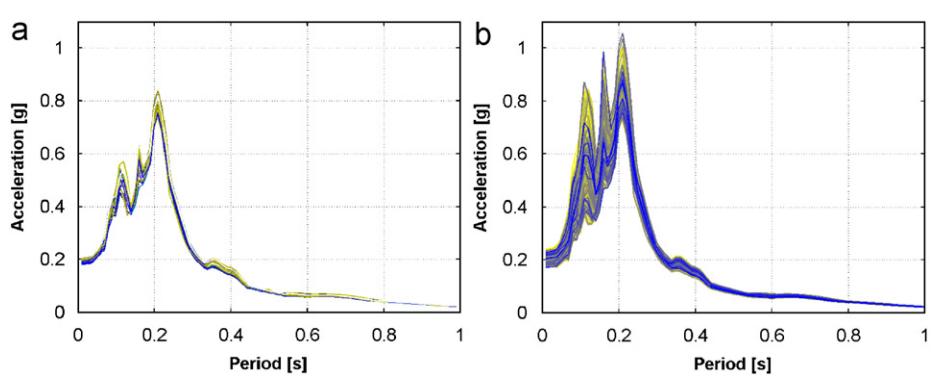

Fig. 12. Seismic site response at La Salle site—response spectra of equivalent profiles: (a) combined active and passive tests and (b) passive tests.

obtained by using only passive data or active-plus-passive data. The reference earthquake for La Salle site is given by a magnitude 5.5 event at epicentral distance 18 km, associated to a normal fault rupture; the PGA for the site is equal to 0.11 g for a return period of 475 years. The reference bedrock has been placed at 100 m from the ground surface on the basis of the information retrieved from a combined reflection–refraction survey [38].

The results obtained in terms of response spectra for the same reference ground motion are presented in Fig. 12, for the two testing conditions considered (respectively, only passive data in Fig. 12a and active-plus-passive data in Fig. 12b). For both datasets, the pattern of the response spectra is quite consistent for all the equivalent velocity profiles obtained by the MC inversion, although the range of variability of the response spectra

associated to the analysis of only passive data is quite high, whereas the results for the combined active–passive data are very satisfactory.

Box-plots are used to depict the statistical distribution of peak amplifications for one of the selected accelerograms for the site in Fig. 13. The higher uncertainty in the determination of the shear-wave velocity profiles reflects also in the peak amplitude determination with a wider dispersion of the results around the median value. However, it can also be observed how the median values for both testing configurations are quite similar. The same pattern can be observed in Fig. 14 that reports the result in terms of PGA.

This observation deserves some further comments. Each profile obtained with MC inversion potentially represents the final result of a deterministic inversion of the experimental data set. Hence the total range of variation covered in Figs. 13 and 14 is an approximation of the possible errors in estimating the site response. The comparison of the results in Figs. 13 and 14 shows that the reliability of analyses based only on passive data is poor.

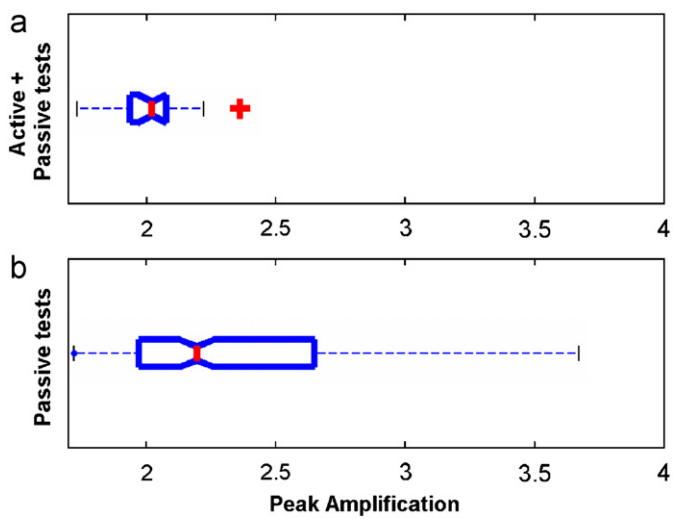

Fig. 13. Seismic site response at La Salle site—box plots of the distributions of the peak amplification for one of the selected accelerograms for combined active and passive tests and passive tests only.

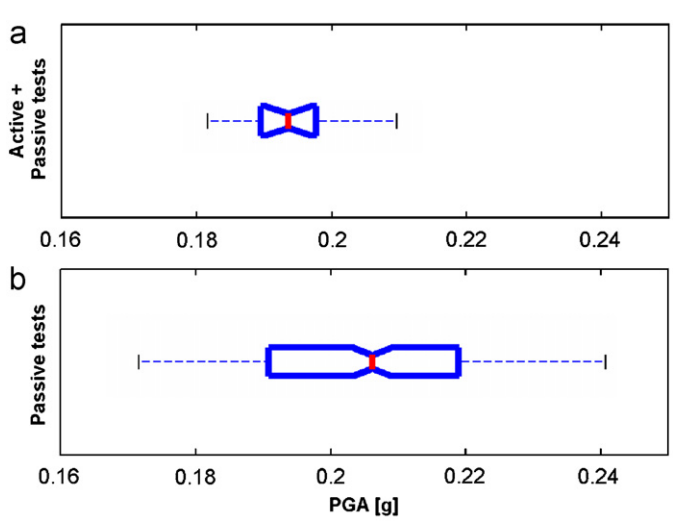

Fig. 14. Seismic site response at La Salle site—box plots of the distributions of the PGA for one of the selected accelerograms for combined active and passive tests and passive tests only.

On the other side, the difference in median values between the two datasets is small. The bottom line is that microtremor analyses are to be considered either in association to active datasets or explicitly accounting for solution non-uniqueness as in the present study.

Coherently with the results of the synthetic tests, in both testing configurations, the uncertainty in the determination of shear-wave velocity profiles is not propagated in the evaluation of local site response parameters corroborating the conclusion drawn from the synthetic example and the previous case history. For example, the high uncertainty in the determination of layer velocity for the case of passive tests alone (average CoV over the profile of the order of 15%) is strongly reduced in the evaluation of the PGA (average CoV of 8%).

Moreover, although a direct comparison with the results of the Torre Pellice test site is not possible, as a rough guide it can be stated that the use of combined active and passive tests, rather than active tests alone, results in a more consistent estimate of the local site response parameters, since the soil profile is more accurately described with depth. Indeed the dispersion around the median values of both peak amplification (see for example the result in Fig. 7 with the one in Fig. 13a) and PGA is more reduced and the wider notches are an indication of the higher reliability of the result. This effect is also depicted in Fig. 15, where the evaluation of the effect of the different input motions is reported in terms of box-plots for the combined active and passive tests.

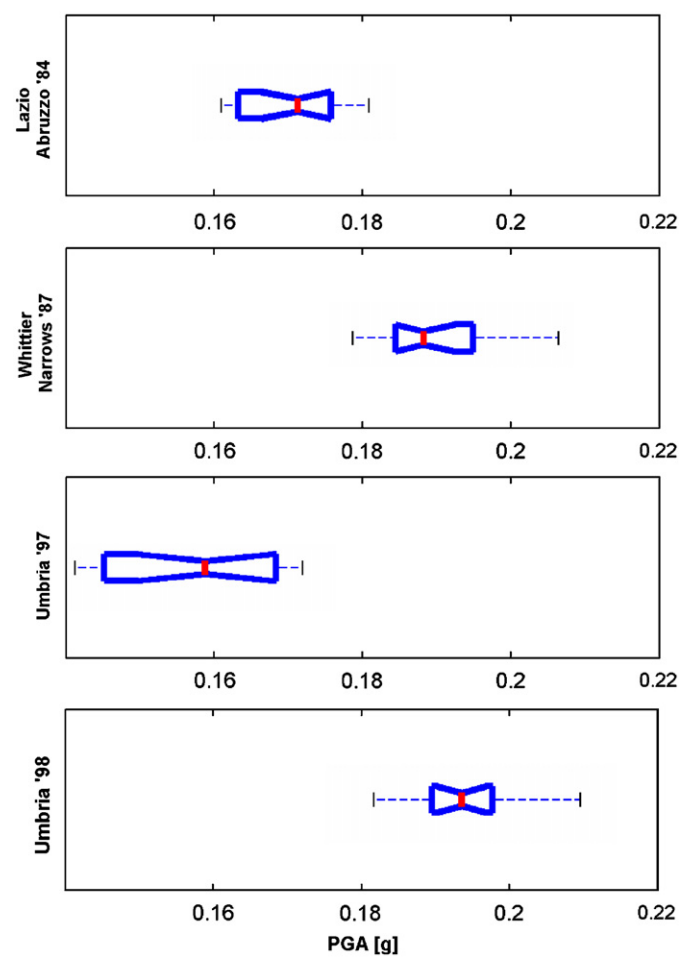

Fig. 15. Seismic site response at La Salle site—box plots of the distributions of PGA for different seismic inputs for combined active and passive tests.

Table 4
La Salle site—PGA values from the four selected accelerograms for combined active and passive test and passive test only.

	PGA (g)					
	Passive tests			Combined active and passive		
	Standard deviation	Mean	Range	Standard deviation	Mean	Range
Whitter Narrows 1987	0.007	0.177	0.039	0.007	0.171	0.020
Lazio-Abruzzo 1984	0.014	0.201	0.080	0.008	0.190	0.028
Umbria 1997	0.011	0.163	0.051	0.011	0.157	0.031
Umbria 1998	0.017	0.205	0.069	0.007	0.194	0.028

Comparing this result with the one in Fig. 8, it is possible to observe a reduced variability in the determination of PGA.

Table 4 reports all of the results of the response analyses in terms of PGA for both testing configurations. There is again a good correspondence in the mean value of the PGA for both cases, but the range (distance between maximum and minimum PGA over the whole set of profiles) and the standard deviations decrease in the case of active-plus-passive data. Also in this situation, as in the previous test site, the effect of the input motion appears to be minor.

6. Conclusions

Non-uniqueness of the solution of the inversion process and the associated uncertainty on the estimation of the shear-wave velocity model are often considered the great pitfall of surface-wave tests. In this study, an assessment of the equivalence problem in surface-wave analysis is presented using both synthetic and experimental data. It is shown that quite different shear-wave velocity models can be obtained starting from the same piece of experimental information. This means that, in principle, if no a priori information is used to constrain the results, errors in the identification of a single parameter can be very high. On the other hand, this study shows how consequences of such uncertainties are very limited for seismic site response studies, at least where a 1D approach is deemed to be reasonable. Indeed, if shear-wave velocity profiles are equivalent with respect to Rayleigh wave dispersion, they are also equivalent with respect to seismic site response. This feature, which can be considered somewhat expected a priori on the basis of the similitude of the underlying soil models, confirms the effectiveness of surface-wave tests in site characterization for seismic response studies.

As far as microtremor analyses (passive methods) are concerned, the consequences of profile uncertainty appear more important, because of the lack of information in the high frequency range. In such conditions an assessment of site response explicitly accounting for non-uniqueness becomes relevant.

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